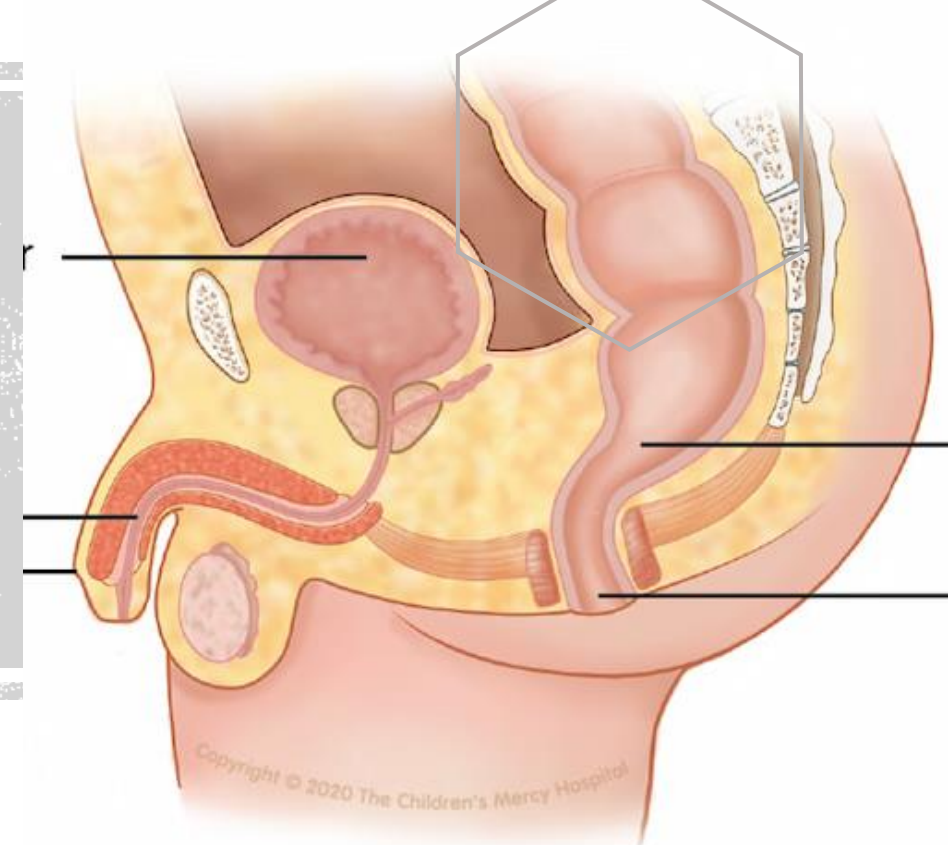


RECTUM AND GENITOURINARY

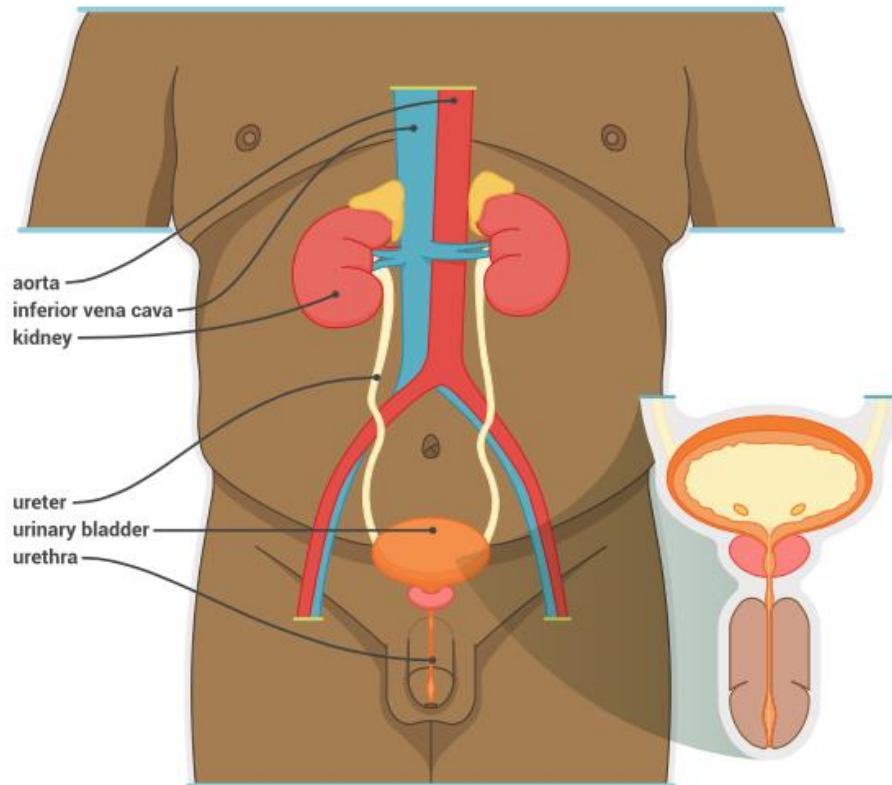
NURS 330

MALE COMMON ANATOMY



COMPREHENSIVE C

URINARY SYSTEM ANATOMY REVIEW



Kidneys

- Fluid and electrolyte balance
- Production of red blood cells
- Filter waste

Ureters

- Move urine from kidneys to bladder

Bladder

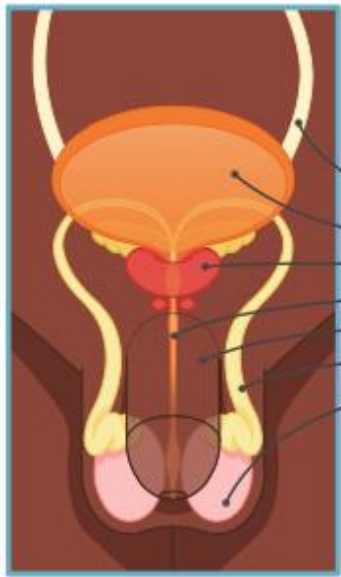
- Holds urine
- Urge to pee around 300mL

Urethra

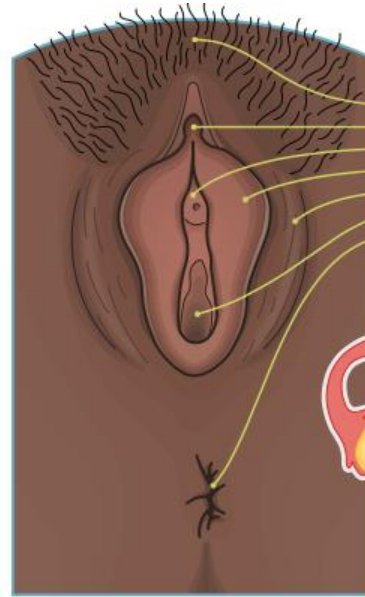
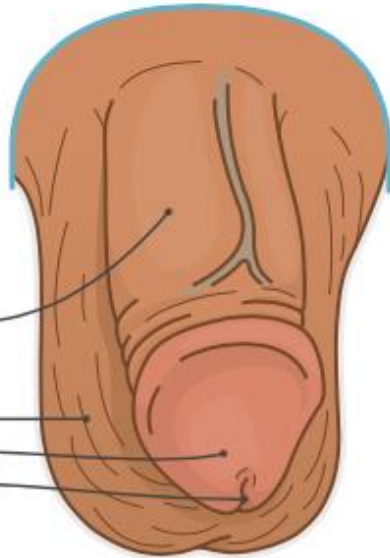
- Passageway for urine and semen

URINARY TERMINOLOGY

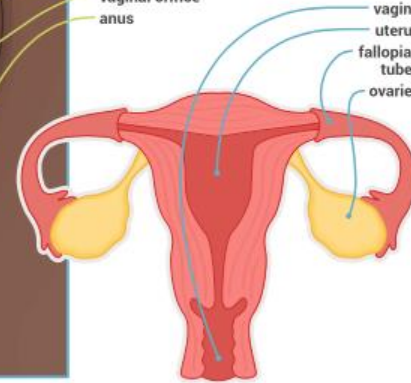
- Retention
 - Inability to empty the bladder
- Hesitancy
 - Trouble starting to urinate
- Urgency
 - Cannot wait to urinate
- Dribbling
 - Loss of urine before or after full stream
- Frequency
 - Occurring more often
- Nocturia
 - Urination throughout the night
- Incontinence
 - Lack of control
- Urge incontinence
 - Loss of muscle causing leaking
- Stress incontinence
 - Loss of control during sneezing, coughing, etc.



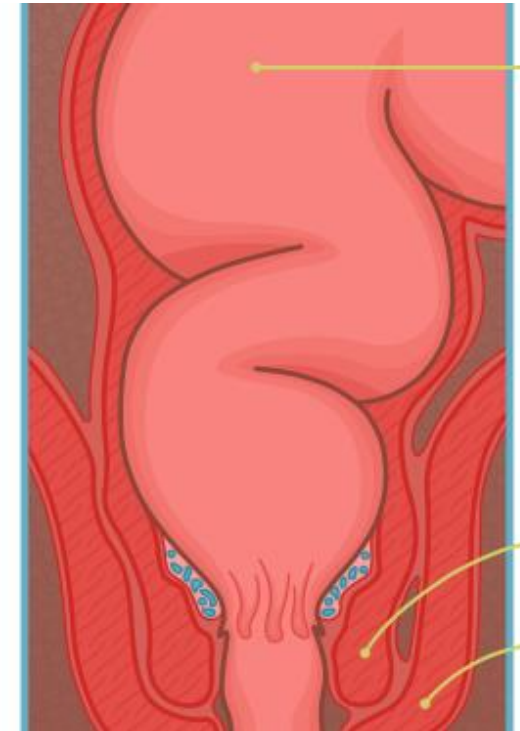
ureter
 urinary bladder
 prostate gland
 urethra
 penile shaft
 vas deferens
 testis
 scrotum
 glans penis
 urethral meatus



mons pubis
 clitoris
 urethral meatus
 labia minora
 labia majora
 vaginal orifice
 anus



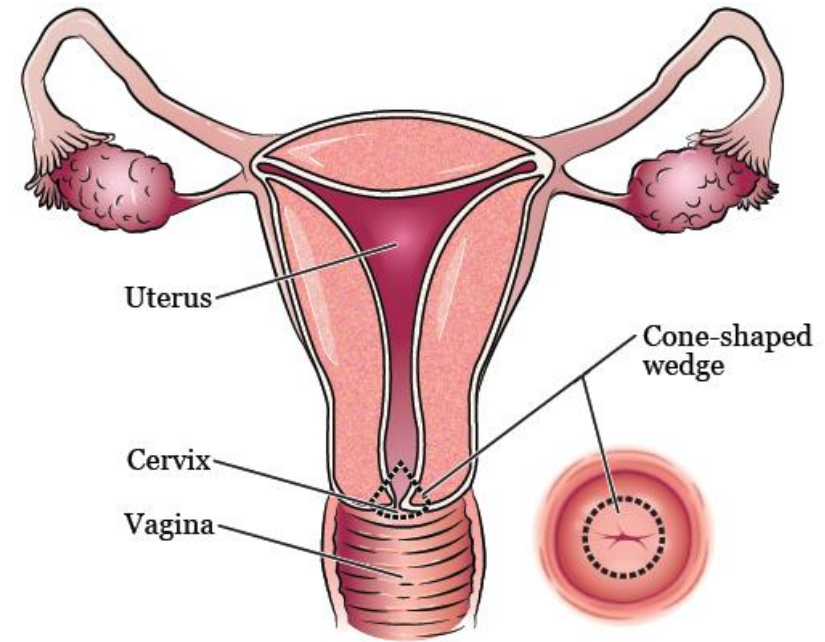
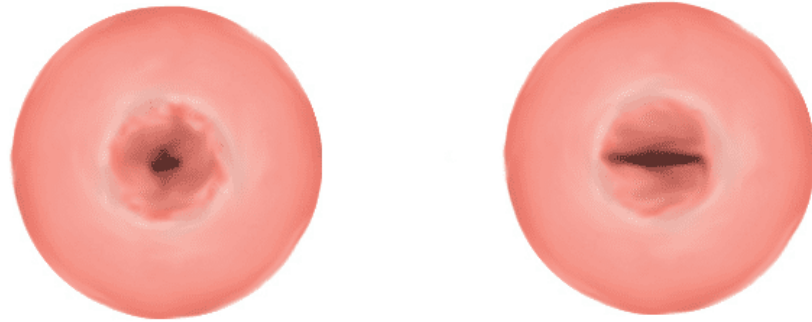
vagina
 uterus
 fallopian tubes
 ovaries



rectum

internal
 sphincter
 muscle
 external
 sphincter
 muscle

FEMALE & MALE REPRODUCTIVE SYSTEM, ANUS



INTERNAL FEMALE GENITALIA: CERVIX

- Lowest portion of uterus
- Produces mucous
- Before childbirth (nulliparous)
 - Smooth with a small circular hole
- After childbirth
 - Hole may be enlarged and more irregular

HEALTH HISTORY

Urinary System

- Any trouble passing urine?
- Urgency to urinate?
- How often do you empty your bladder?
- Any pain?
- Foul odor?

Anus

- Symptoms of hemorrhoids such as itching or pain?
- Change in stools?
- Blood, pus, or mucus?
- Usual bowel routine
- Medication? Laxative?

Reproductive System

- Gender identity and sexual practices nonjudgmentally
- How often do they receive an examination
- When was your last menstrual period?

PROBLEM BASED HEALTH HISTORY

HPV /
vaccines

Ovarian
cancer

Lesions

STIs

Menstrual
history

Menopause
history

Obstetrics
history

Urinary
problems

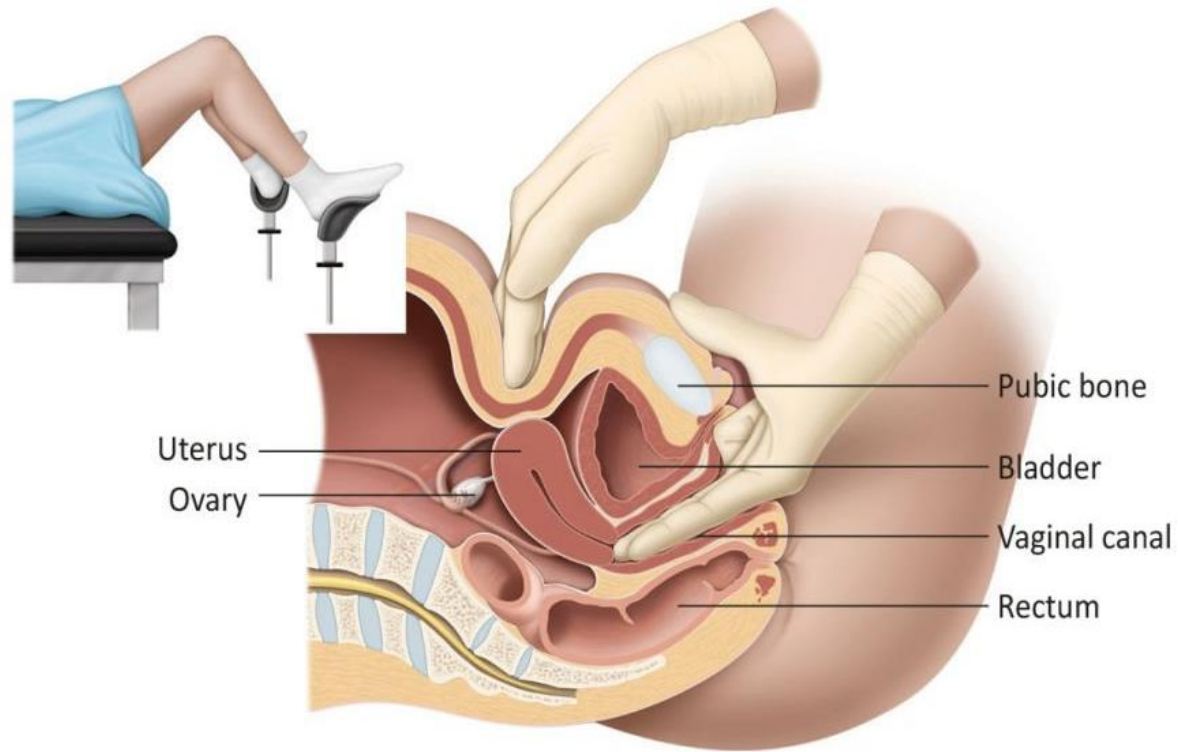
FEMALE GENITOURINARY SYSTEM PREPARATION

- Client should lie supine
 - Knees bent, feet on bed
- Cover the abdomen with sheet
- Knees should drop outward to inspect external genitalia
- Feet may be placed in stirrups
- Ensure client is comfortable
 - Void prior to exam
 - Elevate HOB to allow for eye contact
 - Refrain from legs spreading too wide
 - Explain each step of your exam
 - Use gentle but firm touch



INSPECTION

- Mons Pubis:
 - Note presence of hair and general skin characteristics
- Expected:
 - Hair distribution may vary
 - Should be even with inverted triangle pattern
 - Clear skin with even color
- Variation
 - Soft, straight hair, to coarse and curly
 - Older adult: sparse hair, dry skin
- Unexpected:
 - Swelling / redness
 - Patchy or loss of hair
 - Lesions or ulcerations
 - Pubic lice
- Labia, vestibule, perineum
 - Spread the labia to inspect inner aspects
- Expected:
 - Darker skin color
 - Symmetrical with labia meeting at midline
 - Smooth appearances
- Variations:
 - Scars after vaginal birth
 - Atrophy or dryness after menopause
- Unexpected:
 - Inflammation / ulceration
 - Discolored or malodorous discharge
 - Tenderness
 - White patches (leukoplakia)
 - Lesion, lumps

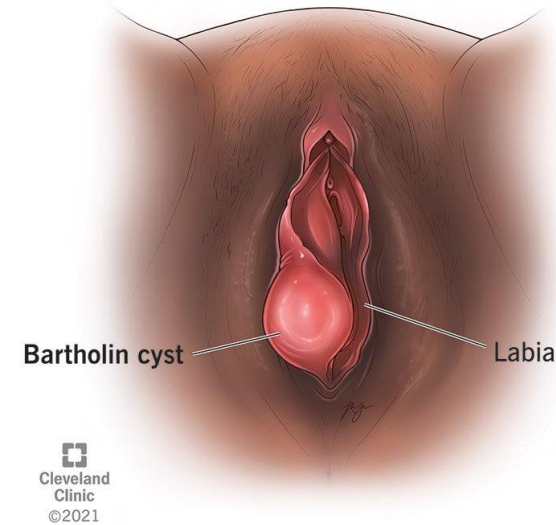
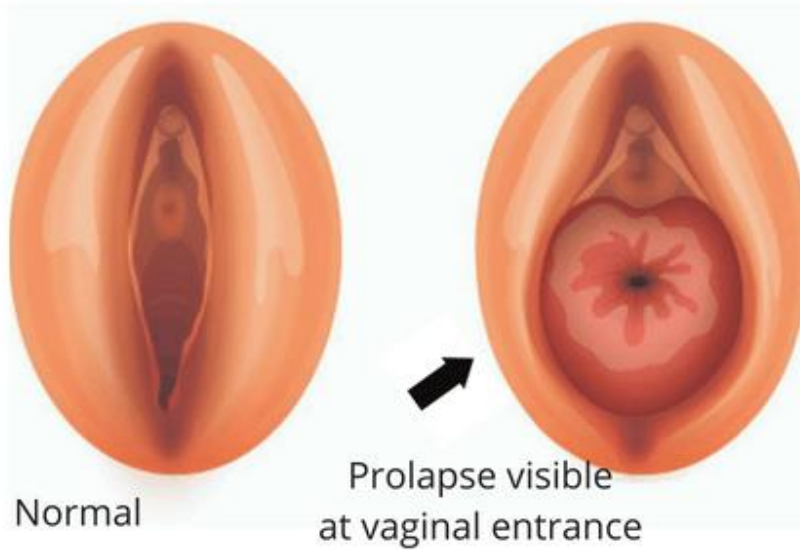


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BIMANUAL EXAMINATION

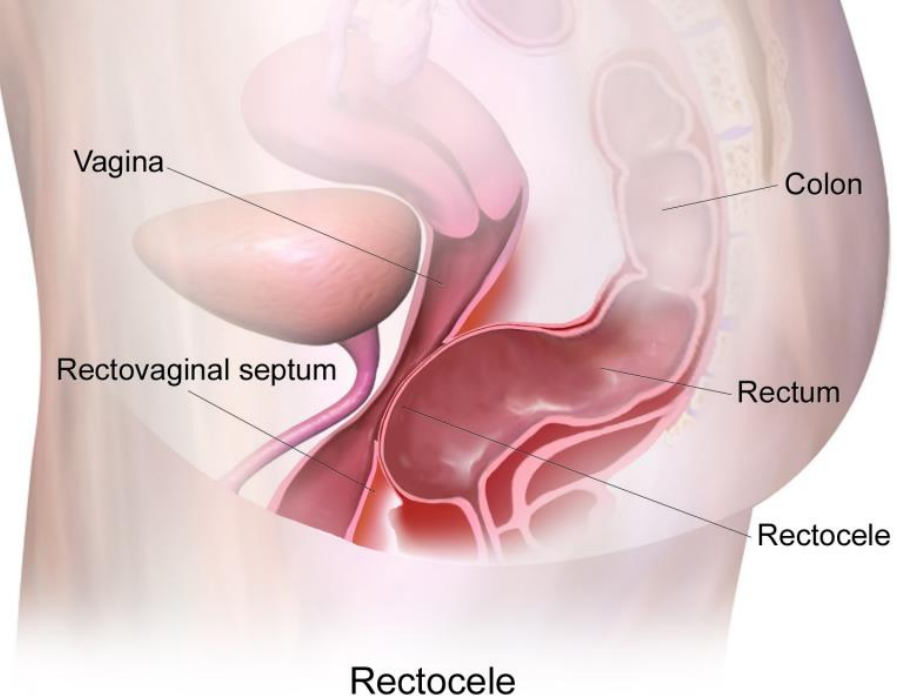
One hand on abdomen,
other inserts two fingers
into vagina

Prolapsed Uterus



UNEXPECTED FINDINGS (CONT'D)

- Cervix at vaginal opening
 - Weakened pelvic muscles causing prolapse of uterus
 - Graded 1-3 based on severity
- Bartholin's gland abscess
 - Fluid-filled pocket causing labia to swell



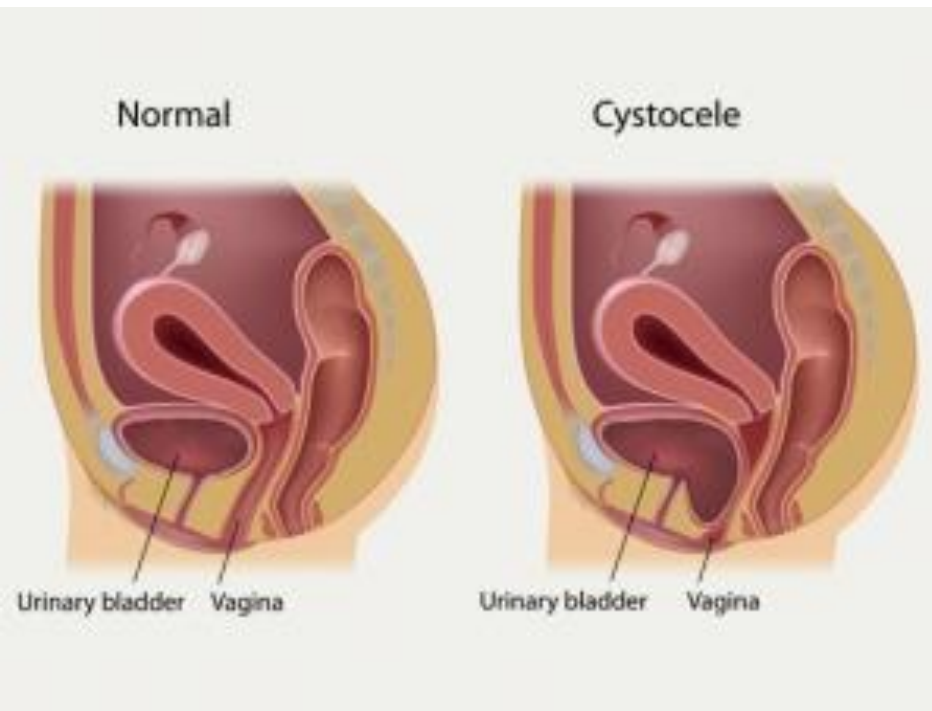
MORE UNEXPECTED FEMALE FINDINGS

Cystocele

- Bladder prolapse into vagina
- Pressure in vagina, stress incontinence

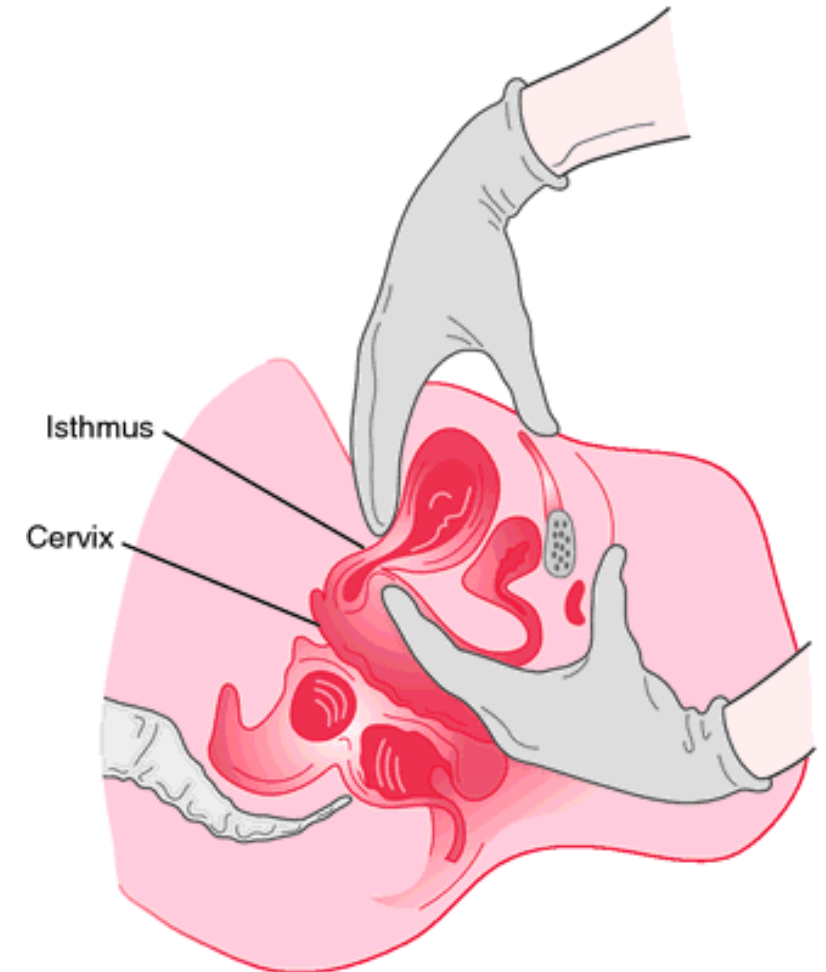
Retrocele

- Rectum prolapse into vagina
- Pressure in vagina, possible constipation



DEVELOPMENTAL: PREGNANCY

- Growing fetus causes symptoms in genitalia
- Increased vascularity and edema of cervix
- Hypertrophy of cervical glands
 - Goodell's sign
 - Softening of the cervix
 - 4 to 6 weeks gestation
 - Chadwick's sign
 - Vaginal mucosa and cervix appear cyanotic
 - 6 to 8 weeks gestation
 - Hegar's sign
 - Isthmus of uterus softens
 - 4 to 6 weeks up to 12 weeks gestation





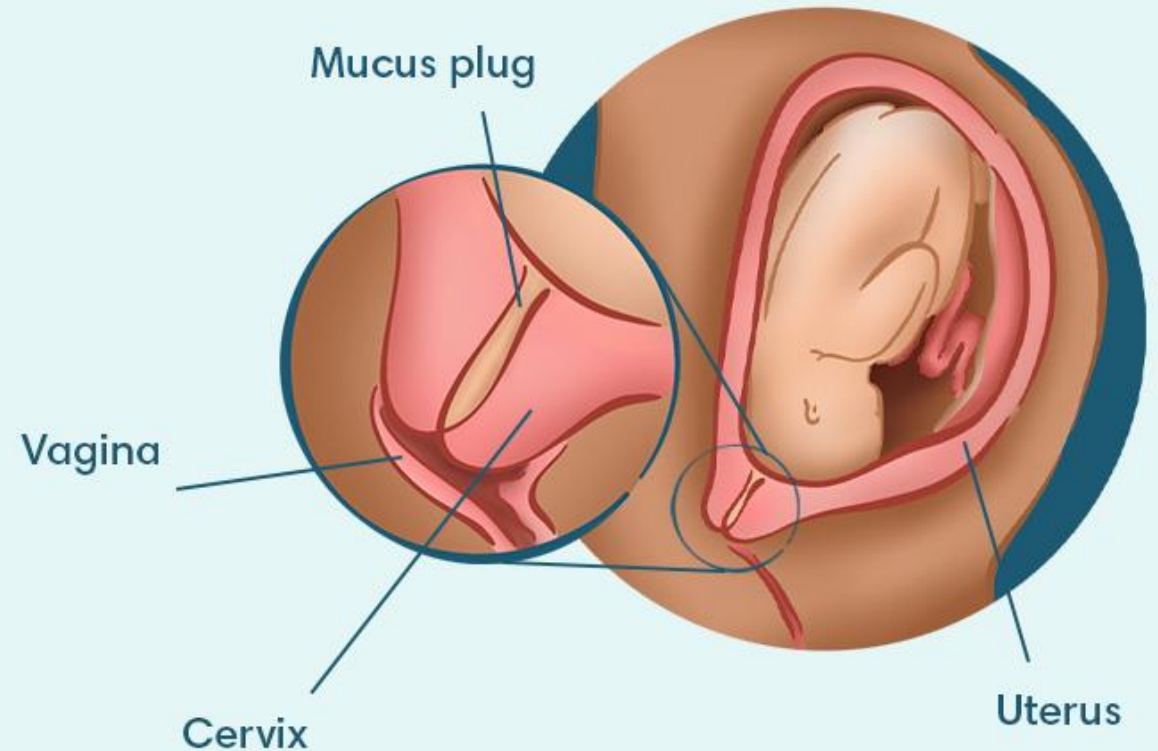
UTERINE CHANGES DURING PREGNANCY

- Capacity increased 500 to 1000 times
 - Hormonal stimulation
 - Increasing content size
- Pushes on bladder causing frequency
- Globular shape by 10 to 12 weeks
 - Moves outside the pelvic area
- Oval shape by 20. to 24 weeks
 - Rises near liver
 - Pushes intestines laterally and superiorly

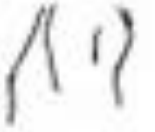
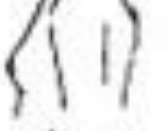
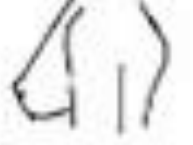
PREGNANCY (CONT'D)

- Mucus plug
 - Thick mucus within the cervical canal
 - Protects fetus from infection
 - Dislodges at end of pregnancy
 - Produces a bloody show
- Cervical and vaginal secretions increase
- Acidic pH by increase of lactic acid
 - Prevents bacterial growth in vagina
 - Increases risk of candidiasis

The Mucus Plug



Source: Cleveland Clinic

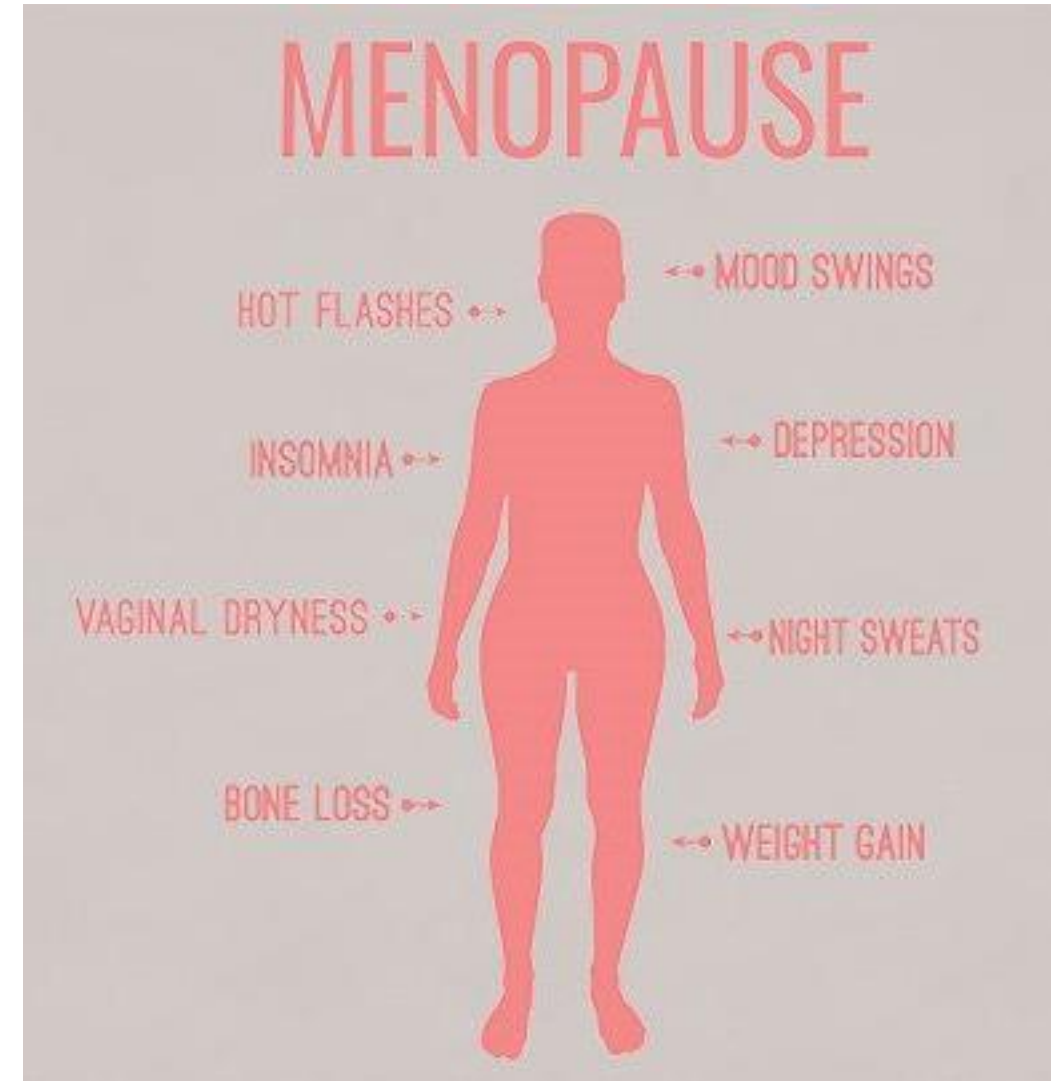
PREPUBERTAL		1		PREPUBERTAL
BREAST BUD		2		PRESEXUAL HAIR
BREAST ELEVATION		3		SEXUAL HAIR
AREOLAR MOUND		4		MID-ESCUTCHEON
ADULT CONTOUR		5		FEMALE ESCUTCHEON

TANNER'S STAGES OF DEVELOPMENT: FEMALE

- Onset of puberty
- Begins between 8 and 12
- Begins with breast budding (thelarche)

DEVELOPMENTAL: AGING WOMAN & MENOPAUSE

- Cessation of menses
- Amenorrhea (absent menses)
- Hormones decrease rapidly
- Occurs between 48 and 51
- Irregular menses will be farther apart and produce lighter flow
- Ovaries no longer produce female hormones

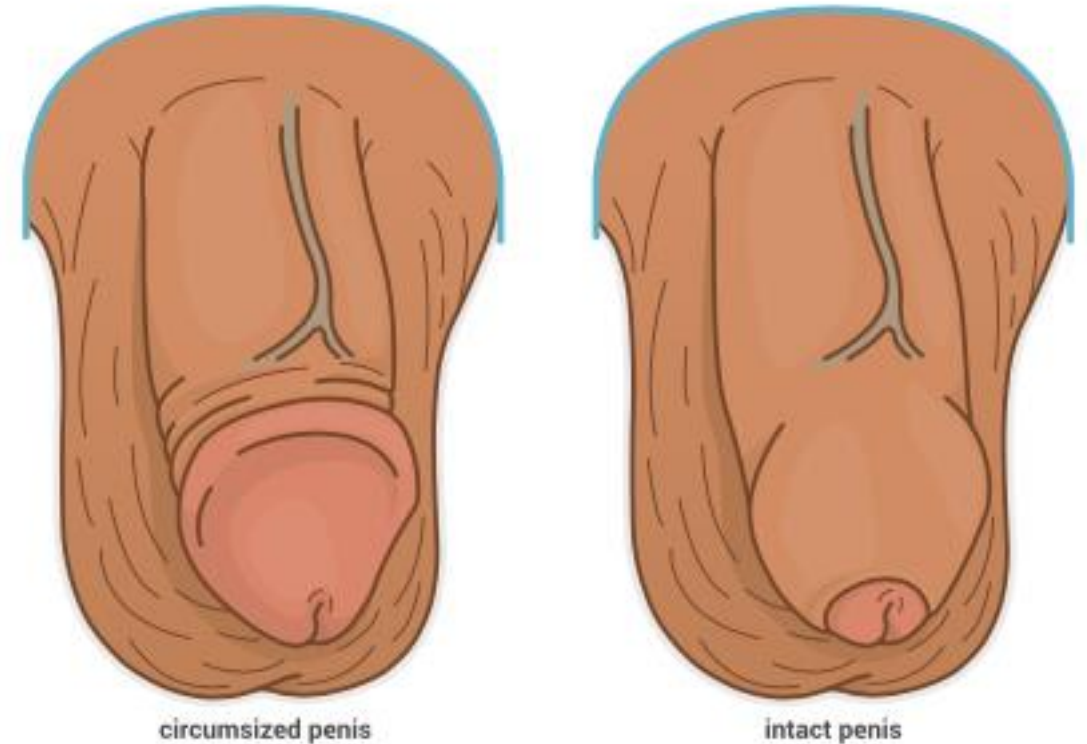


THE AGING WOMAN

- Shrink in uterine, cervix, tissue, and ovary size
 - Ovulation may occur sporadically after menopause
- Droop in uterus due to muscular weakness
 - May cause prolapse
- Vaginal tissue atrophies
 - Sensitive mucosa causing bleeding or inflammation
 - Decreased secretions
 - Alkaline pH

MALE GENITOURINARY SYSTEM ASSESSMENT

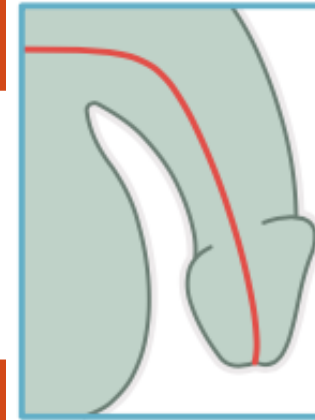
- Client should be supine or standing
- Penis Inspection:
 - Inspect all sides, note hair distribution
 - If foreskin is intact, ask client to retract to inspect glans (tip)
 - If erection occurs, assure client that is normal
- Expected:
 - Skin slightly wrinkled with hair surrounding area
 - Penis should be hairless
 - Large dorsal vein and other vasculature should be present
 - Glans should be smooth
- Unexpected:
 - Inflammation
 - Lesion or ulcers
 - Nodules
 - Pubic lice
 - Phimosis – prepuce cannot retract



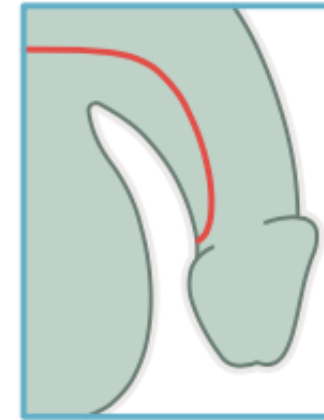
MALE GENITOURINARY SYSTEM (CONT'D)

Urethral meatus

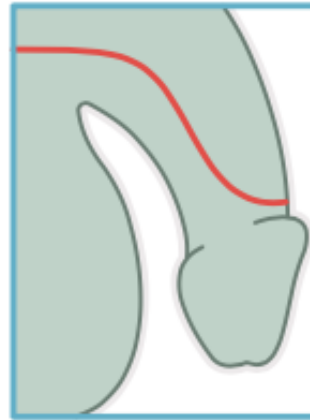
- Midline, surrounding area smooth
- Unexpected: deviation from midline, discharge, redness
- Hypospadias: located on ventral side
- Epispadias: located on dorsal side



normal urethral opening



hypospadias



epispadias

Scrotum, testes

- Darker in skin color with left lower than right
- Equal in size and move freely
- Variations: grow in size during puberty, atrophy in older adults
- Unexpected: edema, nodules, small (<3.5cm), absent

Inguinal & femoral area

- Should be symmetrical & flat
- Unexpected: bulge or swelling (hernia)

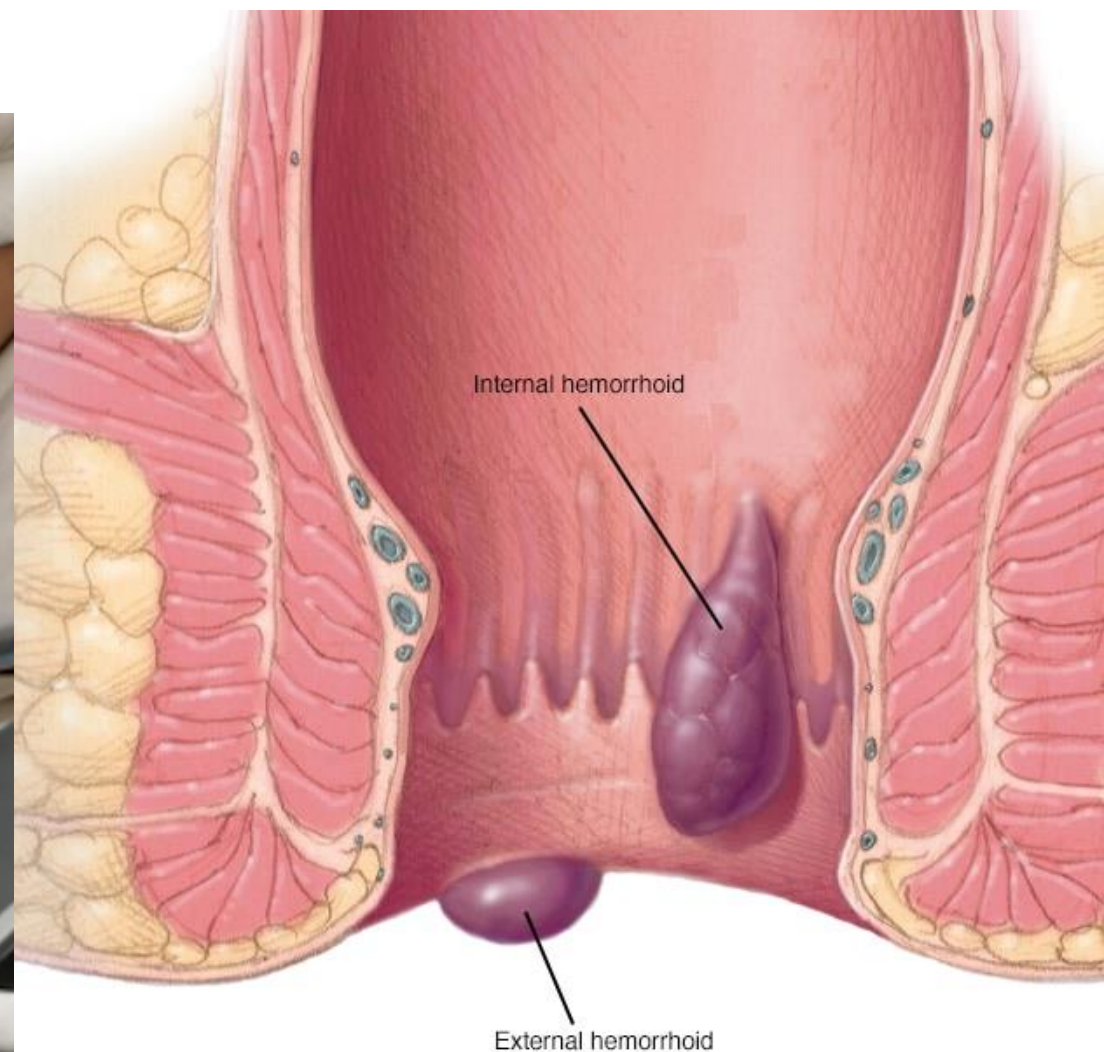
ANAL & PERIANAL ASSESSMENT

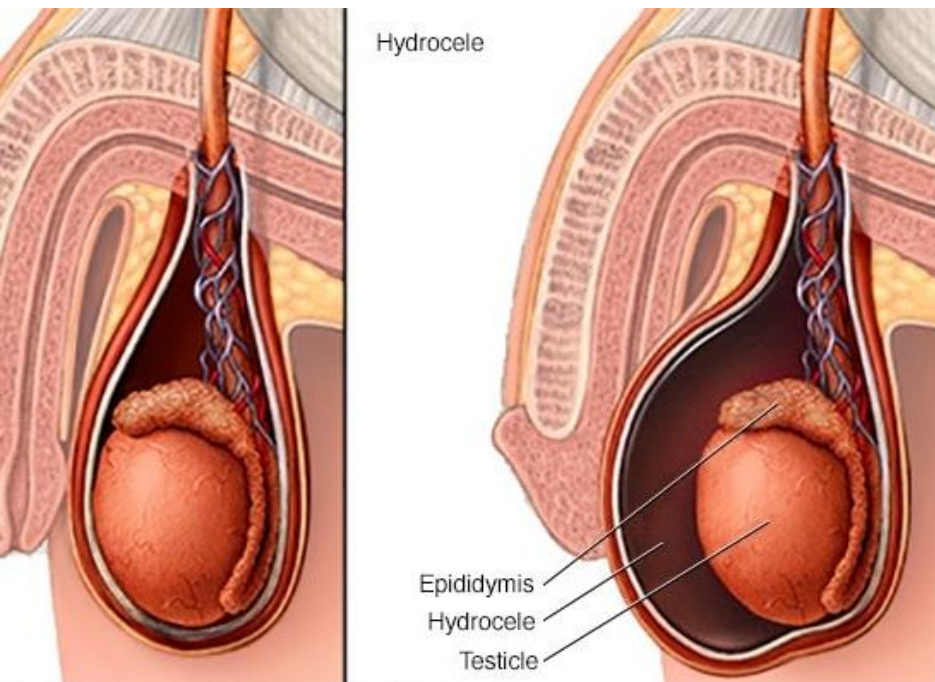
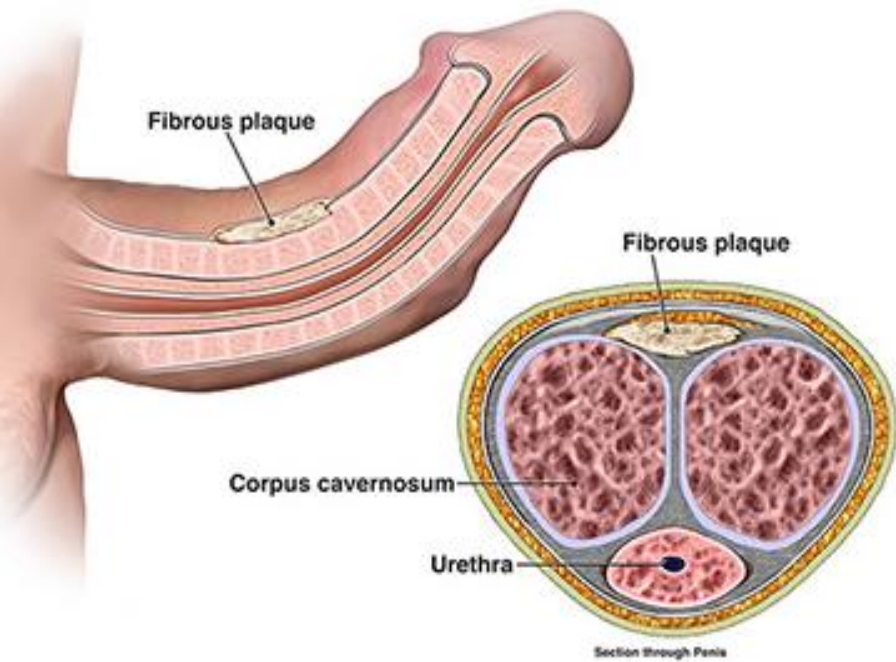
- Position client on their side with hip & knee bent
- Prostate: stand facing the exam table bending forward
- Inspection:
 - Spread the buttocks.
 - Wearing gloves, inspect the color and skin
 - Have client bear down to inspect hemorrhoids, fissures, polyps, tumors, prolapse
- Expected:
 - Skin intact with no protrusions or lesions
 - Darker skin
 - Remain closed on relaxation
- Variations:
 - Decreased control of opening in older adults due to weakened perianal muscles

- Unexpected:
 - Fissure
 - Tear is mucosal wall
 - External hemorrhoids
 - Dilated veins from straining
 - Thrombosed hemorrhoid
 - Bluish-purple, shiny and full
 - Blood clot causes lack of blood flow
 - Rectal prolapse
 - Partial or complete protrusion of rectal wall through anal opening



UNEXPECTED FINDINGS





MORE UNEXPECTED FINDINGS

Hydrocele

- Fluid collection causing testicle swelling
- Rarely needs treatment, usually painless

Peyronie's Disease

- Fibrous plaque growth in penis tissue
- Repeated injury to the penis

SEXUALLY TRANSMITTED DISEASES

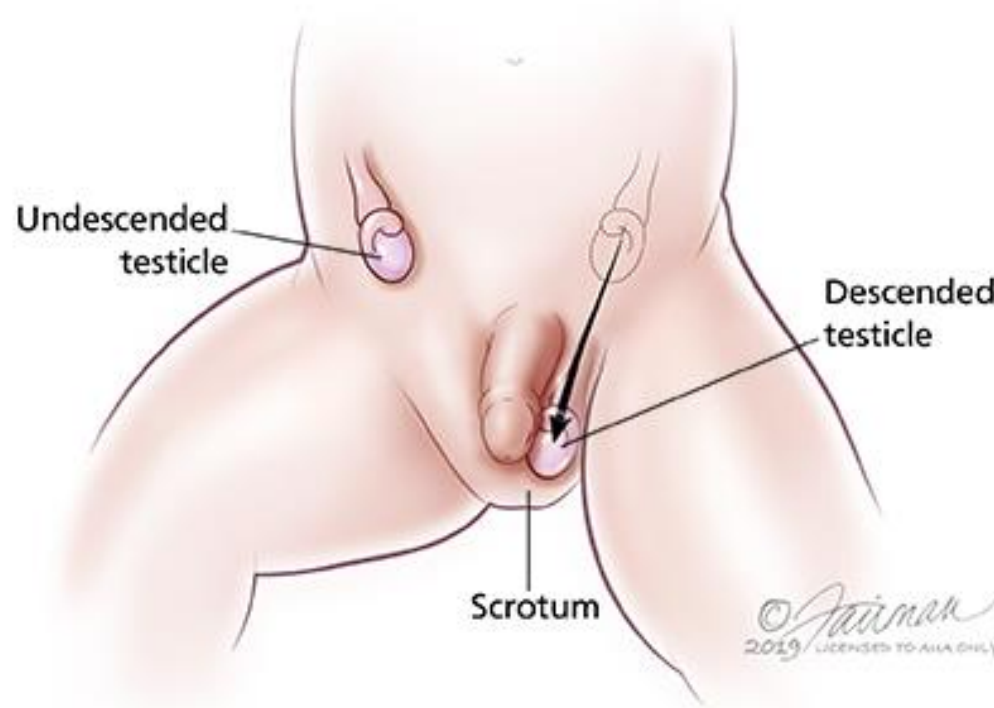
Genital warts:

- Caused by HPV

Herpes simplex virus:

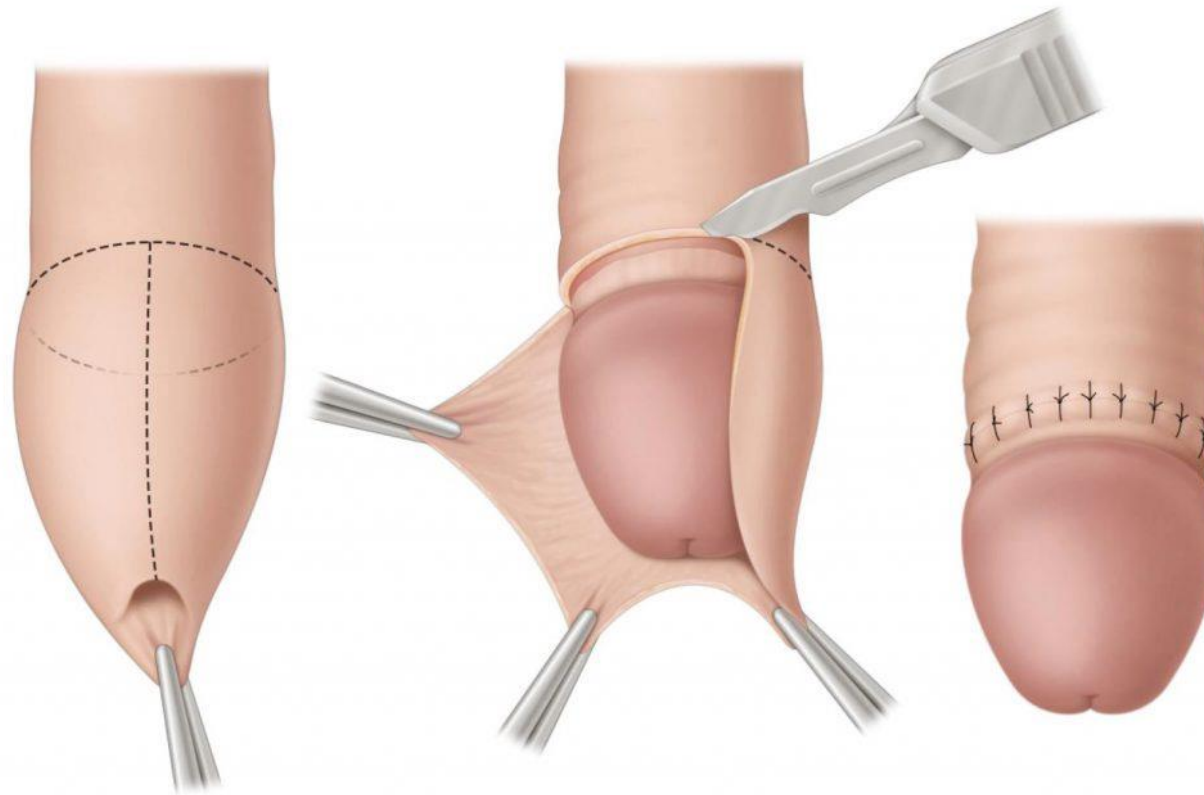
- Shown as vesicles





DEVELOPMENTAL: MALE CHILD

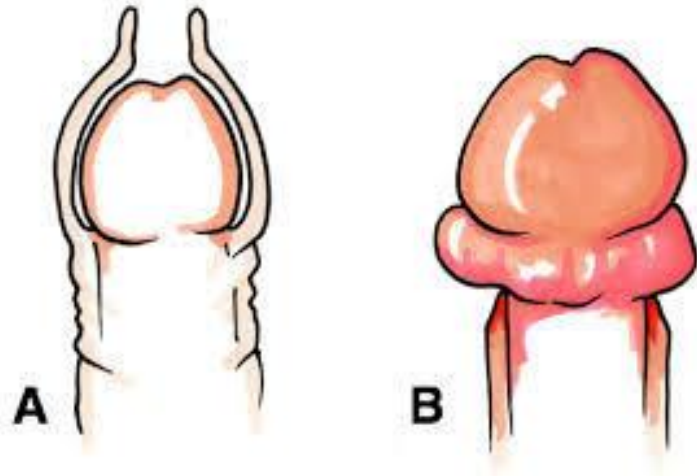
- Testes develop in abdominal cavity
- Testes descent into scrotum before birth
 - Measure 1.5 to 2 cm long, 1 cm wide
- Cryptorchidism
 - Undescended testes



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CIRCUMCISION

- The removal of foreskin
- Understand religious and cultural indications
- Shown to lower risk of STIs, HPV, HIV



FORESKIN & ERECTION CONDITIONS

Phimosis

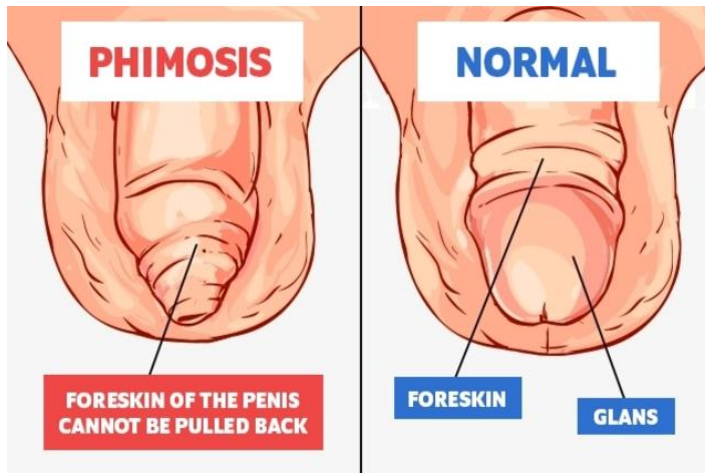
- Foreskin cannot be retracted over the glans

Paraphimosis

- Foreskin becomes stuck behind the glans
- Medical emergency

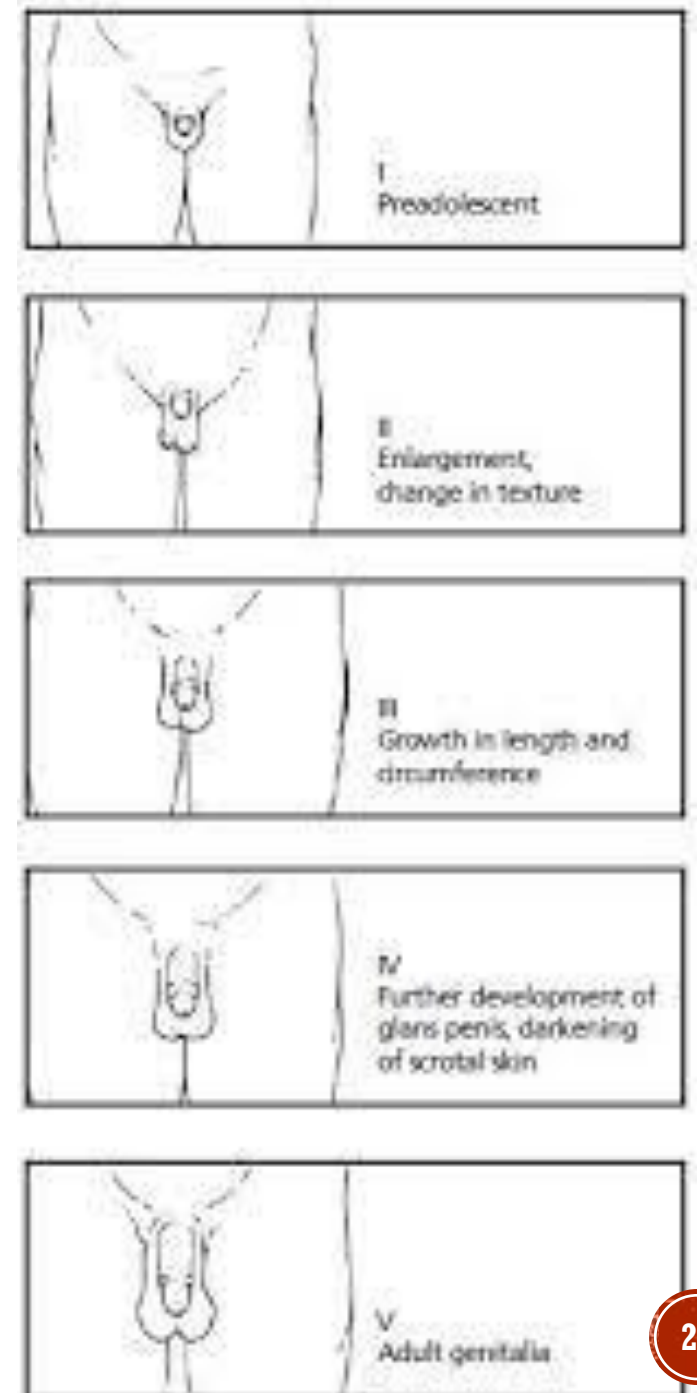
Priapism

- Painful erection not associated with sexual arousal



TANNER'S STAGES OF DEVELOPMENT: MALE

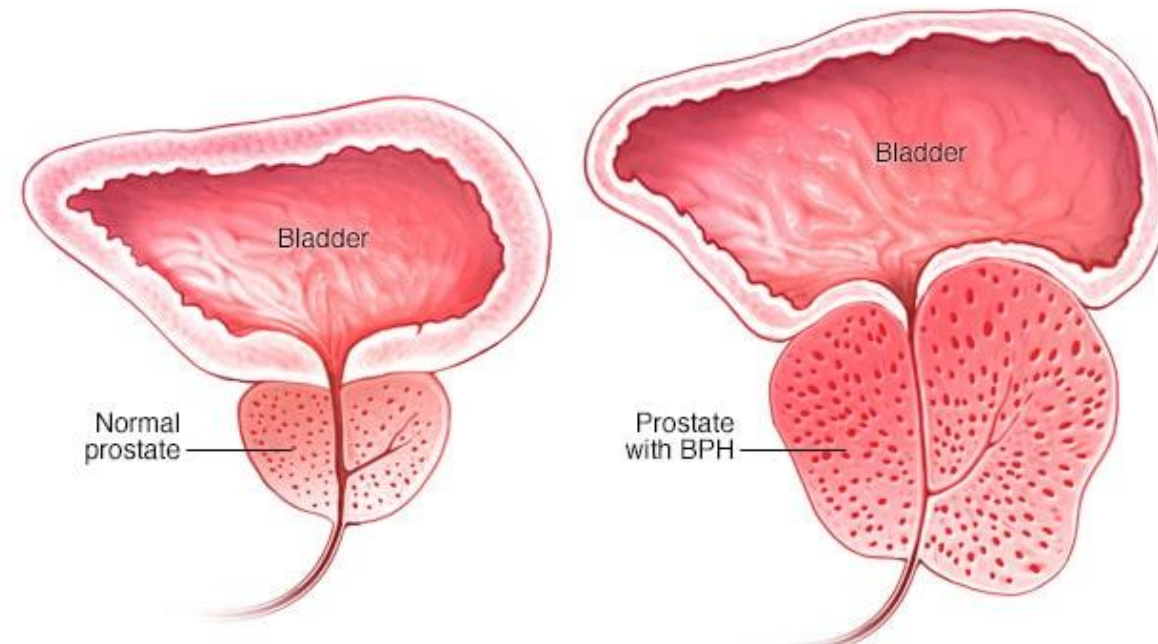
- Onset begins with testicular enlargement
 - Gonadarche
- Typically, between ages 9 to 14



THE AGING MALE

- Production of sperm decreases at 40
 - Will continue until 80s or 90s
- Testosterone production declines after 55 to 60
- Pubic hair decreases
- Penis size decreases
- Scrotum hangs lower
- Testes decrease in size, less firm
- Hypertrophy of prostate
 - May affect urinary system

- Any difficulty urinating?
- Any leaking of urine?
- Do you get up at night to urinate?



URINARY ASSESSMENT

Color

pH &
specific
gravity

BUN

Creatinine

URINARY RETENTION

Acute

- Immediate medical attention
- Complete inability to pass urine
- Painful urge
- Swelling in abdomen
- Catheter is needed

Chronic

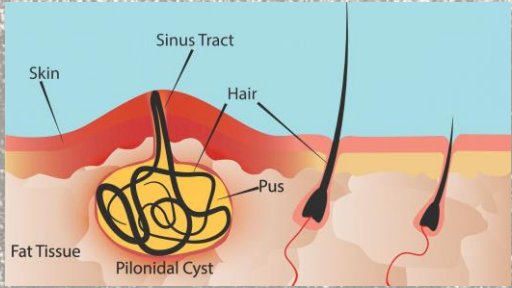
- Urinary frequency
 - > 8 times a day
- Trouble beginning urination
- Intermittent stream
- Urgency after urination

ANAL DEVELOPMENTAL: INFANTS

Meconium: first stool passed within 24-48 hours

Gastrocolic reflex: peristalsis begins after eating





ADDITIONAL ANAL UNEXPECTED FINDINGS

- Pilonidal cyst
 - Cyst with hair located over coccyx
 - May open as a dimple with hair underneath
- Pilonidal dimple
 - Vertebrae not completely closed
 - Not a cause for concern



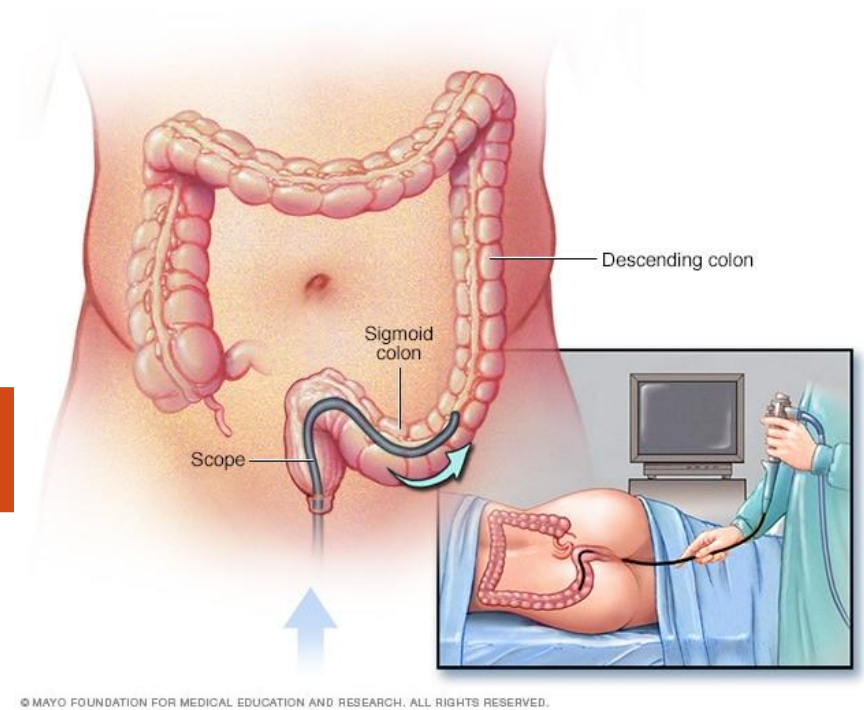
HEALTH PROMOTION

Colorectal cancer screening

- Recommended at 50
- Earlier if family history
- Screening includes:
 - Fecal occult blood once a year
 - Flexible sigmoidoscopy and rectal exam every 5 years
 - Double contrast barium enema every 5 years
 - Colonoscopy every 10 years

Females

- Annual pelvic exam and pap smear at least every 3 between 21 and 65



MALE HEALTH PROMOTION

Prostate & Genitalia

- Annually
- Detects prostate gland lesions
- Common problems: Benign prostatic hyperplasia

Prostate specific antigen, digital rectal exam

- Screens for prostate cancer, PSA should be less than 4
- Completed ages 50 to 70 or 40 to 45 high risk
- Abstain from ejaculation 2 days prior to PSA test

Testicular self exam

- Performed monthly – common in ages 15 to 34
- Annual with provider
- Easy to do after a hot shower
- Should be rubbery and movable
- Lumps may be as small as a pea

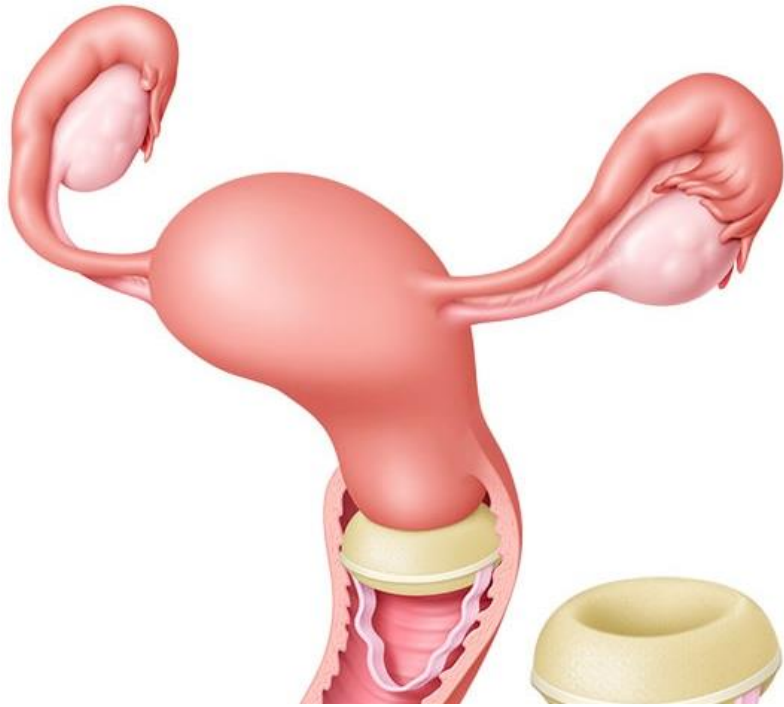
VACCINES

HPV - Gardasil

- Human papilloma virus (HPV) related to cervical and penile cancers
- Spread through skin-to-skin contact
- 3 vaccines over 6 month period
- Should be administered at 11 years old before sexually active

Hepatitis B

- Spread through bodily fluids
- Clients who come in contact should receive vaccine



SAFE SEXUAL PRACTICES

- Understand lifestyle, preferences, culture, and spiritual beliefs
- Contraception
 - Natural
 - Withdrawal, fertility tracking, periodic abstinence
 - Barrier
 - Physical device to prevent sperm from reaching the egg
 - Condoms, diaphragms, sponges
 - Pharmacologic
 - Medication to prevent pregnancy
 - Oral, injectable hormones, vaginal spermicides
 - Surgical
 - Permanent option
 - Vasectomies, tubal ligation



STI PREVENTION & SCREENING

- Education
 - Prevalence of STIs
 - Need for awareness
 - Transmission & risk reduction
- Screening
 - HIV recommended for clients between 15 and 75 or pregnant
 - High risk – test annually
 - Low risk – one time screening
 - Annual screening for chlamydia and gonorrhea if sexually active, younger than 25, multiple partners, may have been exposed



Testing on your own time.

**Getting an HIV and STI
Test is as Easy as 1-2-3.**

-  Complete online forms
-  Go to lab of your choice
-  Get results and follow up

To learn more, visit
testing123ri.com.

COMMUNICATION CONSIDERATIONS: ADOLESCENTS

- Assess sexual behavior and development
- Ask questions appropriate for age
- Ask more questions rather than omit
- Answer all questions they may have
- Avoid sounding judgmental
- Use permission statements:
 - “Often boys or girls your age experience...”
- Use open-ended questions
- Ask about STI protection
- Assess for sexual trauma through conversation

COMMUNICATION: INFANTS & CHILDREN

- Has anyone every touched you between your legs and you did not want them to?
 - They need to know they did nothing wrong
 - "Who are some people you trust?"

Before they turn 18...

1 in 6 boys experience
some form of sexual abuse



1 in 4 girls experience
some form of sexual abuse

